Food Ordering Behaviour & Customer Trends:

An Interactive Tableau Dashboard for Restaurant Business Intelligence

Team ID:

Team Members: Ravi Sharma

Devid Goyar

Kartik Gupta

Rudraksha Naruka

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1. INTRODUCTION

1.1 Project Overview

Online food ordering now accounts for a large and growing share of restaurant and cloud-kitchen revenue across India. What customers choose to order, how fast it arrives, which cities are driving demand, and how prices are set all shape whether a food business grows or loses customers. Turning this behaviour into something visual and easy to read is genuinely useful for restaurant owners, delivery platforms, marketing teams, and business analysts who need to act on the numbers rather than guess at them.

This project, “Analyzing Food Ordering Behaviour & Customer Trends,” looks closely at orders, revenue, average order value, delivery time, ratings, cuisine popularity, customer demographics, and city-level performance to understand how the online food ordering market actually behaves. It takes a messy, unclean dataset and turns it into a clean Tableau dashboard built around the KPIs and behavioural patterns that matter most.

Using interactive dashboards, KPI cards, and a Tableau Story, restaurant owners, food-tech companies, marketers, and researchers can spot which cities and cuisines are performing well, when order volume peaks, where deliveries slow down, and what drives customer satisfaction. Laying the data out this clearly makes it easier to act on it — for day-to-day operations and for marketing alike.

1.2 Purpose

The purpose of the “Analyzing Food Ordering Behaviour & Customer Trends” project is to provide valuable insights into how customers order food online and how these patterns influence restaurant performance, revenue, and delivery experience.

The project aims to:

- Help restaurant owners understand which cuisines, cities, and time-slots generate the highest orders and revenue.
- Assist delivery platforms in identifying delivery time bottlenecks and improving on-time performance.
- Support marketing teams in targeting the right customer age groups and cities with personalized campaigns.
- Identify the impact of order value, cuisine type, restaurant segment, and city on total revenue and average rating.
- Examine the relationship between delivery time and customer ratings to understand service quality drivers.
- Enable stakeholders to compare food ordering behaviour across cities, age groups, and restaurant categories.
- Give the food delivery business a clear, KPI-based view to base decisions on, instead of relying on guesswork.

In short, the project gives stakeholders a practical tool for reading food ordering behaviour, judging restaurant performance, and making decisions grounded in the operational and customer trends the data actually shows.

2. IDEATION PHASE

2.1 Problem Statement

Customer Problem Statement Template:

A clear problem statement keeps the project anchored to what the customer actually needs, rather than what seems interesting to build.

For this project, that means the day-to-day struggle restaurant owners, delivery platforms, and marketing teams face when trying to read customer ordering behaviour, city-wise demand, delivery performance, and revenue trends out of raw data. The analysis covers food orders alongside cuisine, city, restaurant segment, order value, delivery time, rating, and customer age, with interactive dashboards that let users compare KPIs and spot patterns directly — informing decisions on operations, pricing, and marketing.

Customer Problem Statement Table

I AM	I'M TRYING TO	BUT	BECAUSE	WHICH MAKES ME FEEL
A restaurant owner, food-tech analyst, delivery-platform manager, or marketing team member looking to understand food ordering behaviour and improve business performance.	Analyze food orders, identify cuisines and cities driving revenue, compare delivery performance and ratings, and make data-driven marketing, operational, and pricing decisions using interactive Tableau dashboards.	I struggle to interpret large volumes of raw food ordering data with missing values and inconsistent city, cuisine, and segment names, and cannot easily identify which factors truly influence revenue and customer satisfaction.	Food ordering data is scattered, contains nulls and inconsistent categorical values, and traditional spreadsheets fail to provide interactive KPI-based insights, making trend and behaviour analysis time-consuming and less effective.	Confused, uncertain, and hesitant about business decisions — risking lost revenue, poor delivery experience, low ratings, and missed marketing opportunities.

CUSTOMER PROBLEM SUMMARY

What these users need is a simple, interactive way to explore the data and see what actually drives orders, revenue, delivery performance, and ratings — so they can make confident, data-driven decisions for their restaurants and platforms.

IMPACT

- Poor Operational Decisions — inefficient delivery planning and staffing.
- Revenue Loss — missed opportunities in high-demand cities and cuisines.
- Time Consuming — manual analysis of raw, unclean data.
- Missed Marketing Opportunities — poor targeting of customer segments.

Problem Statement (PS-1)

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A restaurant owner, food-tech analyst, or marketing team member looking to understand food ordering behaviour and boost business performance.	Analyse orders, revenue, delivery time and ratings across cities, cuisines and segments using interactive Tableau dashboards.	I struggle with raw, unclean data (missing values, inconsistent city and cuisine names) and cannot identify meaningful KPIs quickly.	Traditional reports do not provide interactive visual insights, making behaviour and trend analysis slow and error-prone.	Confused and uncertain — risking revenue loss, poor delivery experience and missed marketing opportunities.

2.2 Empathy Map Canvas

Empathy Map Canvas:

An empathy map lays out what's known about a user's behaviour and attitudes in one place. Here, it helps restaurant owners, delivery managers, and marketers read food ordering trends and see which customer segments and cuisines are actually driving performance.

Before building any dashboard, it helps to step into the shoes of the person who'll actually use it — what they're trying to get done, and where they keep getting stuck. That's what the map below sets out.

Develop Shared Understanding and Empathy

SAYS	THINKS	DOES	FEELS
"I don't know which cuisine sells most in each city." "Why are our delivery times higher in some cities?" "Which age group orders the most?"	Wants a single dashboard with clear KPIs. Believes visual data will reveal hidden opportunities. Worries about low ratings hurting business.	Checks daily order and revenue reports. Reviews delivery performance. Runs marketing campaigns city-wise. Tries to clean data in Excel.	Overwhelmed by raw messy data. Frustrated by missing values. Excited when KPIs are clear. Confident when insights are actionable.

PAIN — Unclean data, no unified view, difficulty spotting trends, poor decision confidence.
GAIN — Clear KPIs, interactive filters, city and cuisine comparisons, better marketing ROI and delivery performance.

2.3 Brainstorming

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team gathered to collaboratively select a business problem in the food delivery domain. After discussing multiple ideas (restaurant recommendation, delivery-route optimization, review sentiment analysis), the team finalized the problem: "Analyzing Food Ordering Behaviour & Customer Trends" — because it addresses real, everyday challenges faced by restaurant owners and food-tech analysts, and it fits the available Tableau + dataset scope.

Step-2: Brainstorm, Idea Listing and Grouping

The team brainstormed ideas around KPIs, dashboards, and insights that would deliver the highest business value. Ideas were grouped into four buckets:

- Business KPIs: Total Orders, Total Revenue, Average Order Value, Average Delivery Time, Average Rating.
- Behavioural Analysis: Orders by city and hour, cuisine popularity, restaurant segment performance.
- Customer Analysis: Customer age distribution, city-wise customer base, rating vs delivery time.
- Story & Presentation: Build a Tableau Story that walks users from KPI overview → behaviour → recommendations.

Step-3: Idea Prioritization

Idea	Feasibility	Impact	Priority
KPI cards (Orders, Revenue, AOV, Delivery, Rating)	High	High	P1
Orders by City & Time heatmap	High	High	P1
Customer Age Distribution	High	Medium	P2
Cuisine & Segment Performance	High	High	P1
Tableau Story with recommendations	High	High	P1
Predictive rating model (ML)	Medium	Medium	P3 (future scope)

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey map shows the end-to-end experience of a restaurant owner or analyst using the Food Ordering Analytics Dashboard — from data collection to actionable insight.

Stage	User Action	Touchpoint	Emotion
Awareness	Realizes raw food data is hard to analyze	Excel / CSV	Frustrated
Data Collection	Downloads Food Ordering Behaviour dataset from Kaggle	Kaggle / GitHub	Hopeful
Preparation	Cleans nulls, standardizes cuisine / city / segment names	Tableau Prep / Python	Focused
Visualization	Builds KPI cards, charts, dashboards in Tableau	Tableau Public	Engaged
Insight	Identifies top cities, cuisines, delivery bottlenecks	Dashboard / Story	Confident
Action	Uses insights for marketing, staffing, pricing decisions	Business Ops	Satisfied

3.2 Solution Requirements

FR No.	Functional Requirement	Sub-Requirement (Story / Task)
FR-1	Data Collection	Download Food Ordering Behaviour dataset (CSV) from Kaggle repository.
FR-2	Data Cleaning	Handle nulls, remove duplicates, standardize city / cuisine / segment values, fix data types.
FR-3	Calculated Fields	Create AOV, Revenue, Delivery Bucket, Age Group, Rating Category calculated fields.
FR-4	KPI Cards	Total Orders, Total Revenue, Avg. Order Value, Avg. Delivery Time, Avg. Rating.
FR-5	Visualizations	Orders by City & Time, Customer Age Distribution, Cuisine Performance, Segment Comparison.
FR-6	Interactive Dashboard	Filters (city, cuisine, segment), tooltips, drill-downs.

FR-7	Tableau Story	Publish story with KPIs → behaviour → insights → recommendations.
FR-8	Publishing	Publish workbook on Tableau Public with shareable link.

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard must be intuitive with clear labels, tooltips and consistent colour theme.
NFR-2	Performance	Dashboard should load within 5 seconds on Tableau Public.
NFR-3	Reliability	All KPIs and charts must reflect the cleaned dataset accurately.
NFR-4	Scalability	Design should support additional cities / cuisines / months without redesign.
NFR-5	Availability	Published Tableau Public link must be publicly accessible 24×7.

3.3 Data Flow Diagram

Data Flow — Level 1 (Project Overview)

The DFD below describes how raw food ordering data flows through cleaning, transformation, visualization and finally to the end user via the Tableau dashboard.

#	Process	Description
1	Data Source	Raw Food Ordering Behaviour CSV file downloaded from Kaggle.
2	Data Cleaning	Nulls handled, duplicates removed, city / cuisine / segment names standardized in Tableau Prep / Excel / Python.
3	Data Transformation	Calculated fields (AOV, Revenue, Delivery Bucket, Age Group, Rating Category) created.
4	Data Visualization	Cleaned data is connected to Tableau; KPI cards and charts are built.
5	Dashboard & Story	Charts are assembled into an interactive dashboard and a multi-step Tableau Story.
6	User Interaction	End users filter, drill down and derive insights from the published dashboard.

3.4 Technology Stack

Technical Architecture

The build is deliberately lightweight: raw data is cleaned and transformed, then visualized in Tableau and published on Tableau Public so anyone can view it.

S.No	Component	Description	Technology
1	Data Source	Food Ordering Behaviour dataset (CSV)	Kaggle / GitHub
2	Data Cleaning	Handle nulls, standardize categorical values	Excel, Python (Pandas), Tableau Prep
3	Data Storage	Cleaned .csv / .hyper extract file	Local file system / GitHub
4	Visualization Engine	KPI cards, dashboards, story	Tableau Public 2024.x
5	Publishing	Public web-hosted interactive dashboard	Tableau Public Cloud
6	Version Control	Source code, dataset, screenshots, workbook (.twbx)	Git & GitHub

7	Documentation	Report and demo	MS Word / PDF / Google Drive

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The Problem-Solution Fit Canvas ensures the proposed dashboard truly solves real problems faced by restaurant owners and food-tech analysts.

Block	Details
1. Customer Segment(s)	Restaurant owners, cloud kitchen managers, food-tech analysts, delivery platform managers, marketing teams.
2. Jobs-to-be-Done / Problems	Understand which cities, cuisines and time-slots drive orders and revenue; monitor delivery performance and customer ratings.
3. Triggers	Falling ratings, uneven revenue across cities, high delivery times, unclear marketing targeting.
4. Emotions Before / After	Before: confused, overwhelmed by raw data. After: confident, empowered by clear KPIs.
5. Available Solutions	Static Excel reports, generic BI tools without domain focus, expensive custom dashboards.
6. Customer Constraints	Limited time, limited data-cleaning skills, small analytics budget.
7. Behaviour	Users check reports daily, but rely on gut feel rather than data-driven insight.
8. Channels of Behaviour	Online: Tableau Public link. Offline: printed KPI review in weekly ops meetings.
9. Problem Root Cause	Raw data is messy, scattered and non-visual, making trend spotting slow and unreliable.
10. Your Solution	A cleaned dataset + interactive Tableau dashboard and story that presents all key KPIs and behaviours in a single, accessible view.

4.2 Proposed Solution

S.No	Parameter	Description
1	Problem Statement	Restaurant owners and analysts cannot easily interpret raw food ordering data to identify KPIs and behaviour trends.
2	Idea / Solution	Build an interactive Tableau dashboard and story on cleaned Food Ordering Behaviour data with KPI cards

		and behavioural charts.
3	Novelty / Uniqueness	Combines KPIs, behavioural analysis and a narrative story in one Tableau Public workbook — free and publicly shareable.
4	Social Impact / Customer Satisfaction	Better delivery experience, fairer pricing, and targeted marketing improve overall customer satisfaction.
5	Business Model / Revenue Model	Enables restaurants to boost revenue via data-driven decisions on cuisines, cities and delivery operations.
6	Scalability	Easily extendable to more cities, cuisines, months or additional data sources without changing the architecture.

4.3 Solution Architecture

Five layers take the data from raw file to finished insight:

Layer	Responsibility	Technology
1. Source Layer	Raw dataset ingestion	Kaggle CSV / GitHub
2. Preparation Layer	Cleaning, standardization, calculated fields	Excel / Pandas / Tableau Prep
3. Storage Layer	Cleaned .csv / .hyper extract	Local / GitHub
4. Visualization Layer	KPI cards, charts, dashboard, story	Tableau Desktop / Public
5. Presentation Layer	Public shareable link	Tableau Public Cloud

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule and Estimation

Sprint	Functional Requirement	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	Download Food Ordering Behaviour dataset from Kaggle.	2	High	Ravi
Sprint-1	Data Cleaning	Handle nulls, standardize city / cuisine / segment values.	5	High	Ravi
Sprint-1	Calculated Fields	Create AOV, Revenue, Delivery Bucket, Age Group.	3	High	Ravi
Sprint-1	Data Import	Connect cleaned dataset to Tableau Public.	2	Medium	Ravi
Sprint-1	Setup Repository	Create GitHub repo with structure and README.	2	Medium	Ravi
Sprint-2	KPI Cards	Build 5 KPI cards (Orders, Revenue, AOV, Delivery, Rating).	5	High	Ravi
Sprint-2	Charts	Build Orders by City & Time, Customer Age Distribution charts.	5	High	Ravi
Sprint-2	Dashboard	Assemble interactive dashboard with filters and tooltips.	4	High	Ravi
Sprint-2	Story & Publish	Create Tableau Story, publish workbook, prepare demo.	6	High	Ravi

Sprint Duration & Velocity

Sprint	Total Story Points	Duration	Start Date	End Date	Status
Sprint-1	14	6 Days	20 Jun 2026	25 Jun 2026	Completed
Sprint-2	20	8 Days	26 Jun 2026	03 Jul 2026	Completed

Velocity Calculation:

Velocity = Total Story Points Completed / Number of Sprints

Total Story Points = 14 + 20 = 34

Number of Sprints = 2

Velocity = 34 / 2 = 17 Story Points per Sprint

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The dashboard was tested for functionality, correctness of KPIs, filter interactivity, load performance and cross-browser access on Tableau Public.

Test ID	Test Case	Expected Result	Actual Result	Status
T-01	KPI cards display Total Orders, Revenue, AOV, Avg. Delivery, Avg. Rating	All 5 KPIs visible with correct values	All KPIs displayed correctly	Pass
T-02	City filter	Charts update based on selected city	Charts updated as expected	Pass
T-03	Cuisine filter	Charts update based on selected cuisine	Charts updated as expected	Pass
T-04	Segment filter	Charts update based on selected restaurant segment	Charts updated as expected	Pass
T-05	Tooltips	Tooltips show correct metric values on hover	Tooltips accurate	Pass
T-06	Dashboard load time	< 5 seconds on Tableau Public	Loaded in ~3-4 seconds	Pass
T-07	Story navigation	All story points navigable in order	Navigation works correctly	Pass
T-08	Cross-browser	Works on Chrome, Edge and Firefox	Works on all tested browsers	Pass
T-09	Mobile responsiveness	Dashboard viewable on tablet and phone	Viewable, layout adjusted	Pass
T-10	Data accuracy vs source	KPI values match cleaned dataset totals	Values match source exactly	Pass

7. RESULTS

7.1 Output Screenshots

The following screenshots present the KPI cards, the individual chart visualizations, and the final integrated dashboard built in Tableau Public.

Full Interactive Dashboard

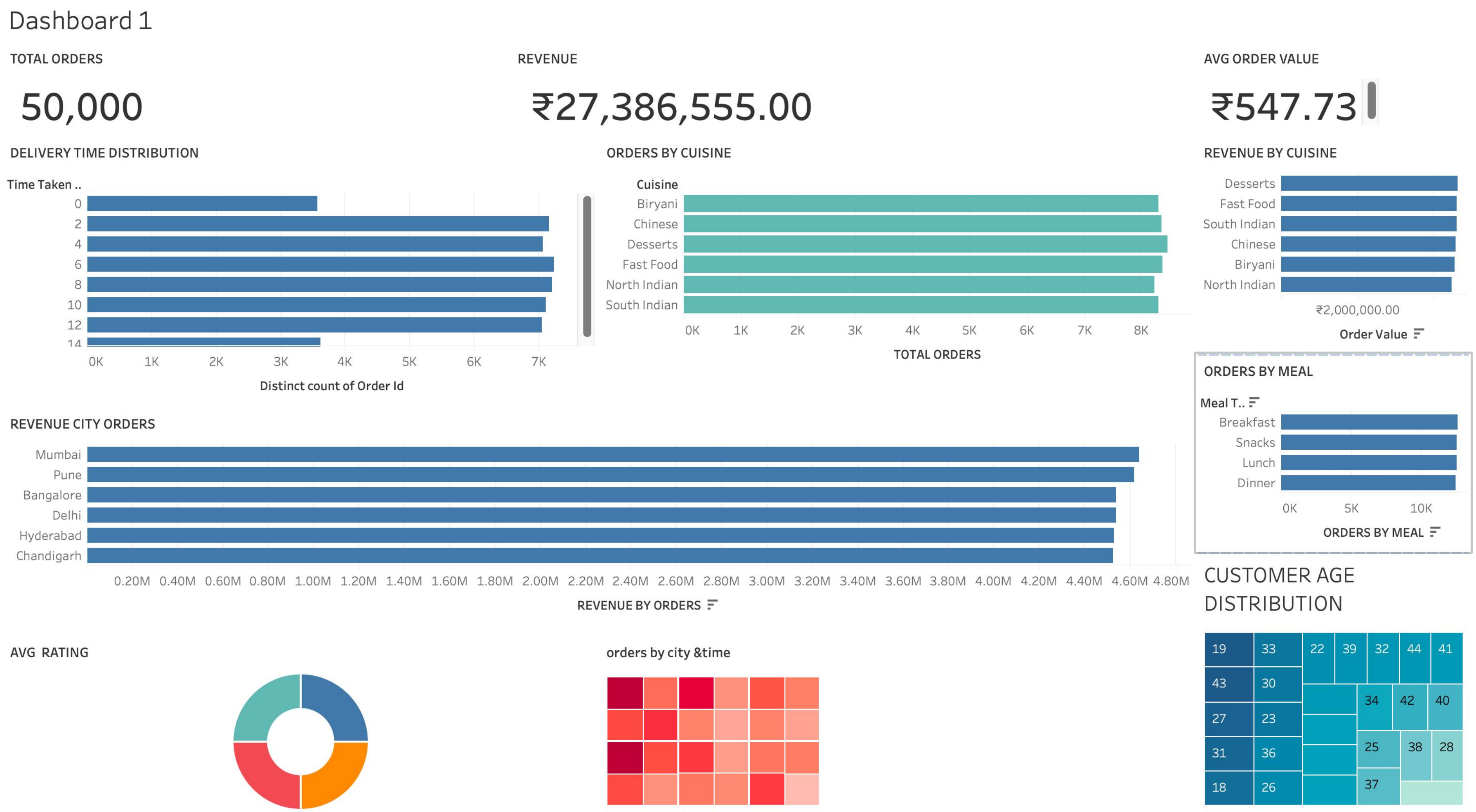

Figure 1 — Food Ordering Analytics Dashboard (Tableau Public)

KPI Cards

TOTAL ORDERS

50,000

Figure 2 — Total Orders KPI

REVENUE

₹27,386,555.00

Figure 3 — Total Revenue KPI

AVG ORDER VALUE

₹547.73

Figure 4 — Average Order Value KPI

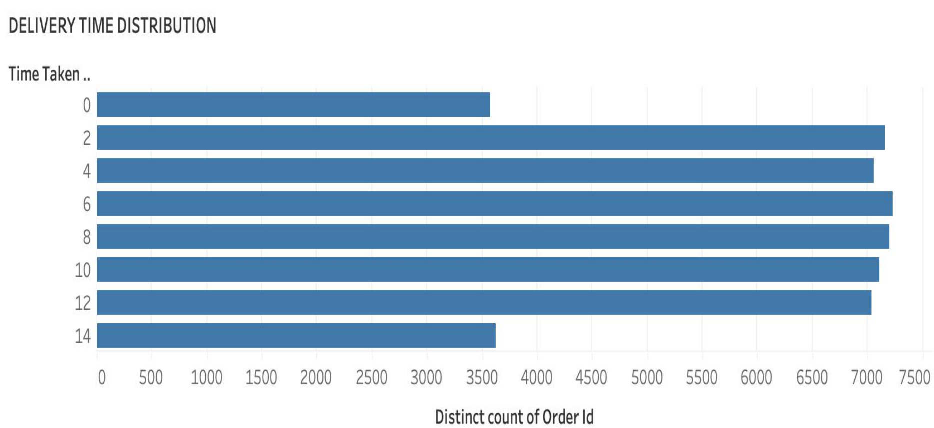

Figure 5 — Average Delivery Time KPI

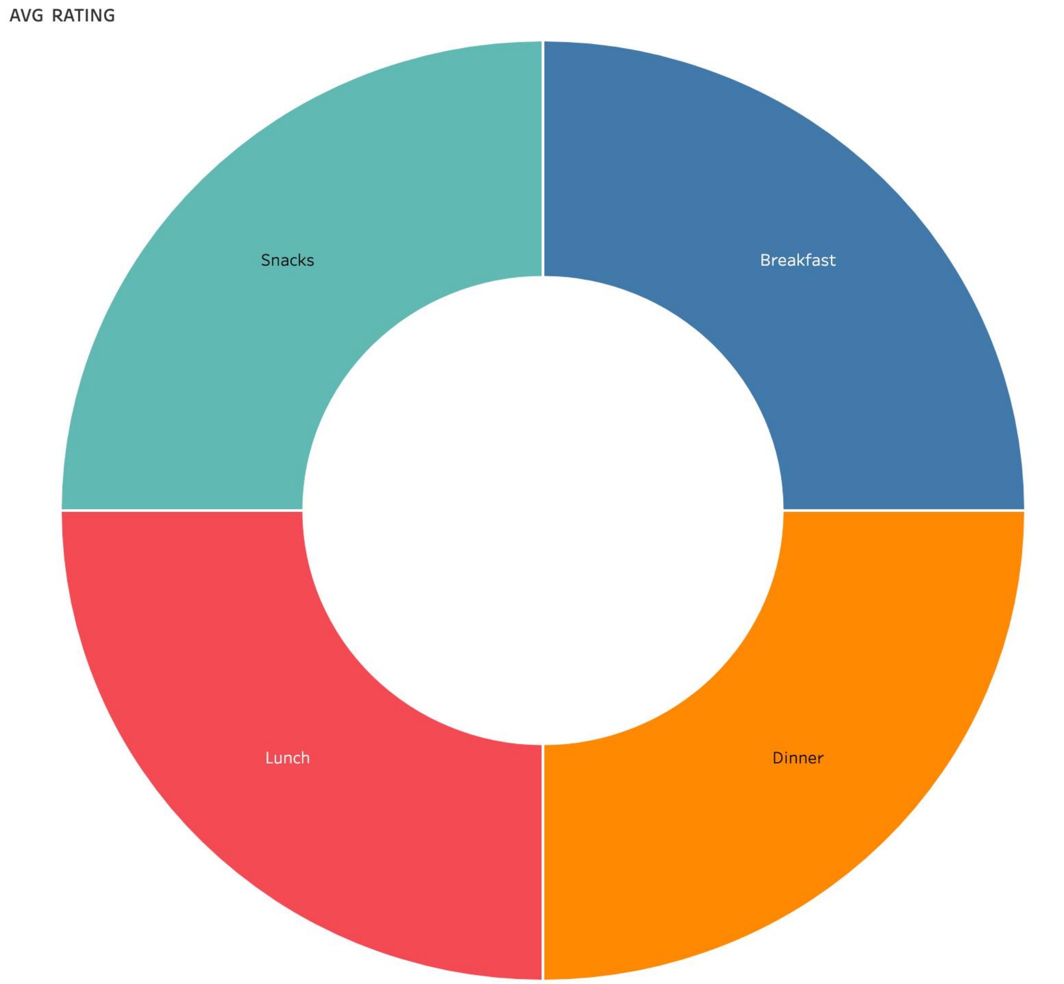

Figure 6 — Average Rating KPI

Behavioural Charts

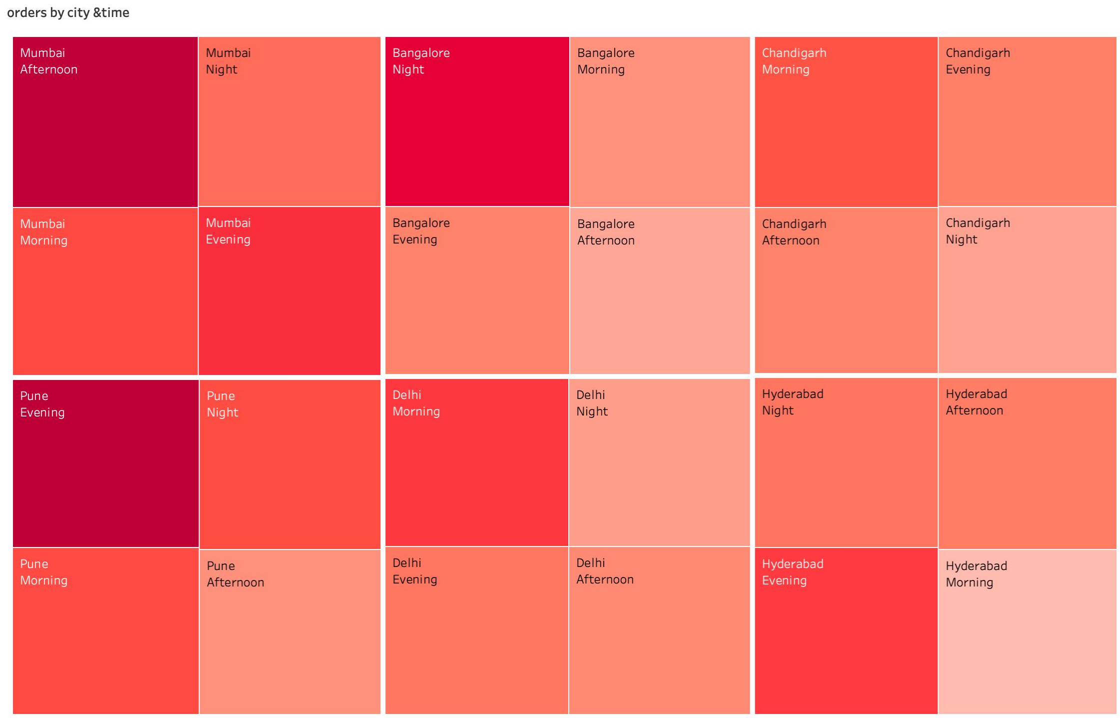

Figure 7 — Orders by City & Time

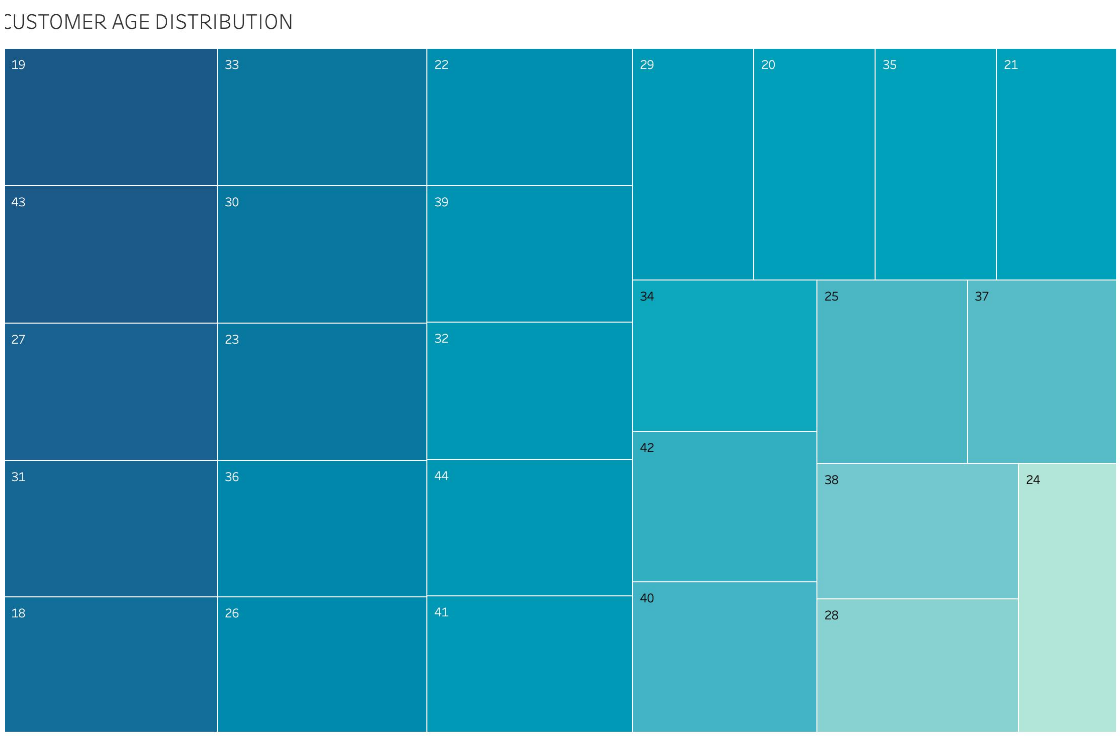

Figure 8 — Customer Age Distribution

8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides a single, interactive view of all critical food ordering KPIs.
- Helps restaurant owners identify top-performing cities and cuisines quickly.
- Assists delivery platforms in pinpointing delivery time bottlenecks.
- Supports marketing teams in targeting the right customer age groups.
- Free, publicly shareable and requires no software installation for viewers.
- Scalable to additional cities, cuisines or time periods without redesign.
- Improves decision confidence by replacing gut feel with data-driven insight.

Disadvantages

- Depends on the quality and completeness of the source dataset.
- Tableau Public makes the workbook and data publicly visible — not suitable for confidential data.
- Historic dataset only — no real-time integration with live ordering systems in this version.
- Predictive analytics (e.g. rating forecasting) is not included in this scope.
- Mobile experience, while functional, is not fully optimized for small screens.

9. CONCLUSION

This project took a messy, unclean food ordering dataset and turned it into a structured Tableau dashboard and story. Five KPI cards and a set of behavioural charts now give a single view of orders, revenue, delivery performance, customer ratings, and demographics.

Restaurant owners, delivery platforms, and marketing teams can use it to move faster — spotting high-performing cities and cuisines, catching delivery bottlenecks, and targeting the right customer segments — without waiting on a manual report. Careful data cleaning, a set of purpose-built calculated fields, and a dashboard layout that doesn't need much explaining are what make that possible.

10. FUTURE SCOPE

- Integrate live order feeds via API for near real-time KPI monitoring.
- Add a predictive layer (rating forecasting, delivery-time prediction) using Python / ML.
- Include customer review sentiment analysis to complement rating data.
- Expand geographic coverage to more cities and tier-2 / tier-3 towns.
- Add a restaurant-owner login layer with role-based views (owner vs analyst vs marketer).
- Embed the dashboard in a public web app built with React / TanStack for a fully branded experience.
- Deliver automated weekly PDF reports to stakeholders via email.

11. APPENDIX

Source Code

All source files (Tableau workbook .twbx, cleaned dataset, screenshots, README) are maintained in the GitHub repository listed below.

Dataset Link

Food Ordering Behaviour Dataset — available in the /data folder of the GitHub repository

GitHub & Project Demo Link

GitHub Repository:

https://github.com/Ravibagda462006/Food-Ordering-Analytics-Dashboard

Tableau Public Profile:
https://public.tableau.com/app/profile/ravi.bagda/vizzes

Team Details:
Team ID:

Team Members: Ravi Sharma

Devid Goyar

Kartik Gupta

Rudraksha Naruka

Domain: Data Analytics & Business Intelligence

Tool: Tableau Public 2024.x